

Hiver SDE Intern Assignment Report

AI Customer Support Agent for AppleSupport

Candidate: Dhanushree

Selected brand: AppleSupport

Dataset: Customer Support on Twitter (TWCS)

1. Problem Framing

This project builds an AI customer-support agent for AppleSupport. The workflow classifies a customer message into one of 11 intents, retrieves relevant historical support evidence, drafts a grounded response, and decides whether the case should be auto-handled or escalated to a human.

The 11 intents are: `ios_update_issues`, `apps_services`, `battery_power`, `device_performance`, `hardware_accessories`, `keyboard_input`, `connectivity`, `photos_data`, `apple_id_account`, `purchases_orders_refunds`, and `other_unclear`.

The design treats the agent as decision support rather than an unrestricted chatbot. Retrieval evidence is intended to reduce unsupported responses, while escalation provides a safety mechanism for ambiguous or sensitive cases.

2. Data Preparation and Labeling

AppleSupport messages were extracted from the TWCS dataset and paired where possible. The processed data contains approximately 106,646 AppleSupport pairs and 105,622 usable messages after cleaning.

The project uses a weakly supervised training set of 21,874 examples, a manually labeled golden intent set with 200 labeled examples plus 100 development examples, and a leakage-safe retrieval corpus containing 5,000 records. Golden examples were excluded from the retrieval corpus, and duplicate customer texts were removed to reduce evaluation leakage.

The intent labels were created using a practical support taxonomy. Because a message may contain multiple issues or insufficient context, `other_unclear` is retained as a fallback category. Weak supervision improves scalability but can introduce label noise.

3. Baselines and Classification Results

Method	Accuracy	Macro-F1	Weighted-F1
Trivial baseline	20.5%	3.09%	6.98%
Rule-based baseline	54.0%	45.26%	51.83%
Zero-shot MiniLM	57.5%	approximately 0.53	—
Weakly supervised classifier	65.5%	approximately 0.53	—

The trivial baseline provides a weak reference point. The rule-based baseline improves performance but is sensitive to vocabulary and overlapping intents. The zero-shot model provides a semantic comparison without task-specific training. The weakly supervised classifier performs better on the evaluated development set, but its labels inherit noise from heuristic rules.

These results are not production-level estimates. The golden set is relatively small, and the classes differ in difficulty and representation.

4. Retrieval, Agent, and Escalation Evaluation

The leakage-safe retrieval evaluation produced a mean Top-1 similarity of approximately 0.7302. A similarity of at least 0.50 was reached by 97.5% of examples, at least 0.60 by 91.0%, and at least 0.65 by 82.0%. Similarity is an evidence-quality signal, not proof of factual correctness.

The end-to-end evaluation produced approximately 55.56% accuracy, 49.13% macro-F1, and 55.35% weighted-F1. Mean retrieval similarity was approximately 0.7322, with standard deviation approximately 0.0925.

A 50-example human escalation audit contained 22 escalation cases and 28 auto-handle cases. The agent achieved 52.0% accuracy, 47.62% precision, 90.91% recall, and 62.5% F1. The false-auto-handle rate was 9.09%. The policy is safety-heavy: it catches many escalation cases but also over-escalates, which may increase human workload.

5. Reply Quality and Judge Agreement

Forty responses were evaluated by a human reviewer:

Criterion	Result
Correctness	4.15 / 5
Relevance	4.97 / 5
Groundedness	4.62 / 5
Completeness	3.45 / 5
Tone	5.00 / 5
Hallucination rate	12.5%
Escalation appropriate	25.0%

The responses were generally relevant, grounded, and polite. Completeness was weaker, showing that responses sometimes omitted useful troubleshooting steps.

An LLM-as-judge evaluation was run using a Groq-hosted model. Agreement was strongest for relevance, tone, and hallucination, and weaker for correctness, completeness, and escalation appropriateness. A second pass by the same human reviewer produced 100% intra-rater agreement and Cohen's kappa of 1.000. This is intra-rater agreement, not inter-annotator agreement. The LLM judge is supplementary and is not treated as ground truth.

6. Top Five Failure Modes

- Hardware issue classified as connectivity:** A customer reported repeatedly cracked phones, but the system predicted connectivity with low confidence and produced a response that may assume AppleCare+ or warranty context.
- Camera-quality complaint classified as update issue:** A grainy camera complaint on a 2016 MacBook Pro was predicted as `ios_update_issues` rather than `hardware_accessories`.
- Keyboard autocorrect issue classified as update issue:** An autocorrect problem was misclassified because the model focused on update-related wording instead of the keyboard symptom.
- Purchased book unavailable for download:** An iBooks access problem was classified as `ios_update_issues` instead of `purchases_orders_refunds`.
- Connectivity problem not escalated:** A customer reported Wi-Fi and Bluetooth dropping after an update. The model predicted `device_performance` and did not escalate, creating a meaningful safety concern.

7. What Is Misleading About My Headline Number?

A single accuracy number can be misleading. The evaluation set is limited, the taxonomy is project-specific, and weakly supervised labels contain heuristic noise. Accuracy also hides class imbalance and does not reveal whether errors occur in high-risk support situations.

Retrieval similarity can also be over-interpreted. Similarity indicates semantic closeness, not factual correctness, successful resolution, or customer satisfaction. A fluent response may still omit an important step or make an unsupported assumption.

The escalation results require similar caution. High recall is useful for safety, but over-escalation increases workload. A low false-auto-handle rate does not prove that every automatically handled case is safe because the audit is small and human escalation judgments can be subjective.

8. Decisions, Limitations, and Next-Week Plan

Major decisions included selecting AppleSupport, defining the 11-intent taxonomy, using weak supervision, creating a golden set, preventing retrieval leakage, implementing two baselines, using sentence embeddings, and grounding generation in retrieved evidence.

Limitations include the small golden set, multi-intent messages, noisy weak labels, imperfect retrieval, a short human escalation rubric, disagreement between human and LLM judges, and the current safety-heavy escalation policy. The evaluation does not yet measure resolution rate, customer satisfaction, or time saved.

Next-week priorities are:

1. Expand the golden set across all intents and edge cases.
2. Use independent annotators and adjudication rules.
3. Add hard-negative training examples and confidence calibration.
4. Introduce structured escalation reasons and severity levels.
5. Improve retrieval with intent-aware filtering and reranking.
6. Add an evidence checker before generation.
7. Evaluate completeness with issue-specific checklists.
8. Add regression tests for observed failure modes.
9. Measure latency, cost, escalation workload, and resolution proxies.

Reproducibility and Safety

The repository contains preprocessing scripts, evaluation artifacts, golden-label files, retrieval outputs, the decision log, and failure analysis. The raw dataset is excluded from Git because of its size and must be downloaded separately using the README instructions.

API keys must be supplied through environment variables and must never be committed. Reported metrics should be reproduced in a clean environment using the documented setup and evaluation commands in README.md.